

```
In [14]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv(r"C:\Users\STUDENTS\Downloads\healthcare_expenses - healthcare_expenses.csv")
df.tail()
```

```
Out[14]:
```

	Patient_ID	Gender	Treatment_Type	Medical_Expense
26	P027	Male	Surgery	1236
27	P028	Female	Diagnostic	1558
28	P029	Male	Therapy	4896
29	P030	Male	Surgery	9999
30	P031	Female	Diagnostic	8794

```
In [3]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 31 entries, 0 to 30
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Patient_ID      31 non-null    str    
1   Gender          31 non-null    str    
2   Treatment_Type  31 non-null    str    
3   Medical_Expense 31 non-null    int64  
dtypes: int64(1), str(3)
memory usage: 1.1 KB
```

```
In [4]: df.isnull().sum()
```

```
Out[4]: Patient_ID      0
Gender      0
Treatment_Type  0
Medical_Expense  0
dtype: int64
```

```
In [5]: df.describe()
```

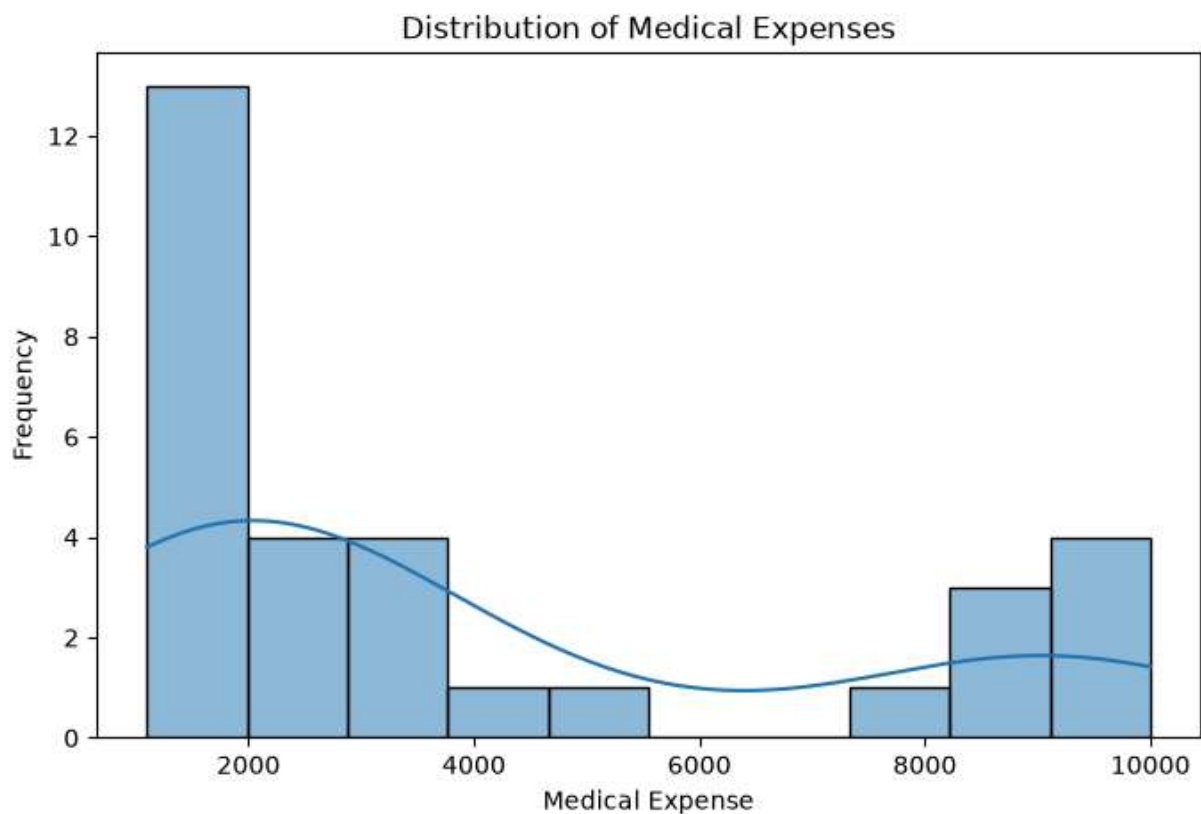
Out[5]:

Medical_Expense	
count	31.000000
mean	3986.000000
std	3181.315273
min	1100.000000
25%	1573.000000
50%	2500.000000
75%	6348.000000
max	9999.000000

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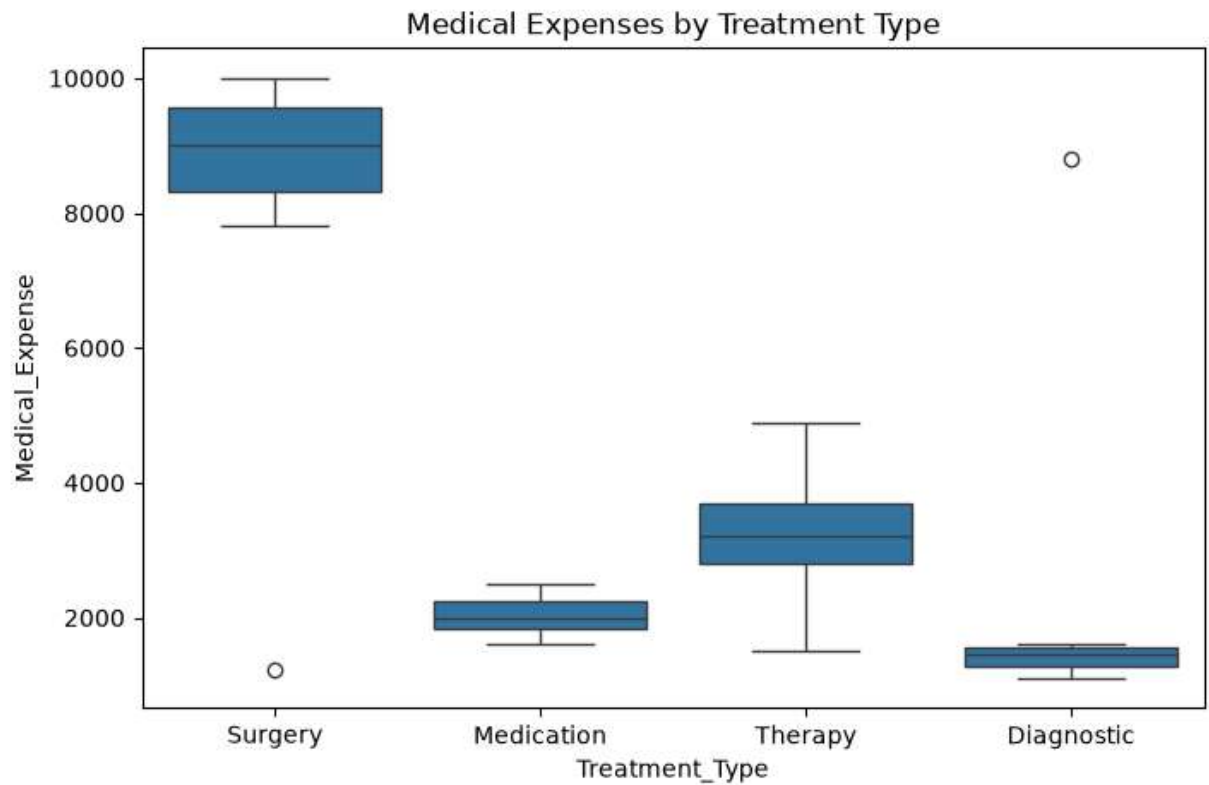
```
In [15]: plt.figure(figsize=(8,5))
sns.histplot(df["Medical_Expense"], bins=10, kde=True)

plt.title("Distribution of Medical Expenses")
plt.xlabel("Medical Expense")
plt.ylabel("Frequency")
plt.show()
```



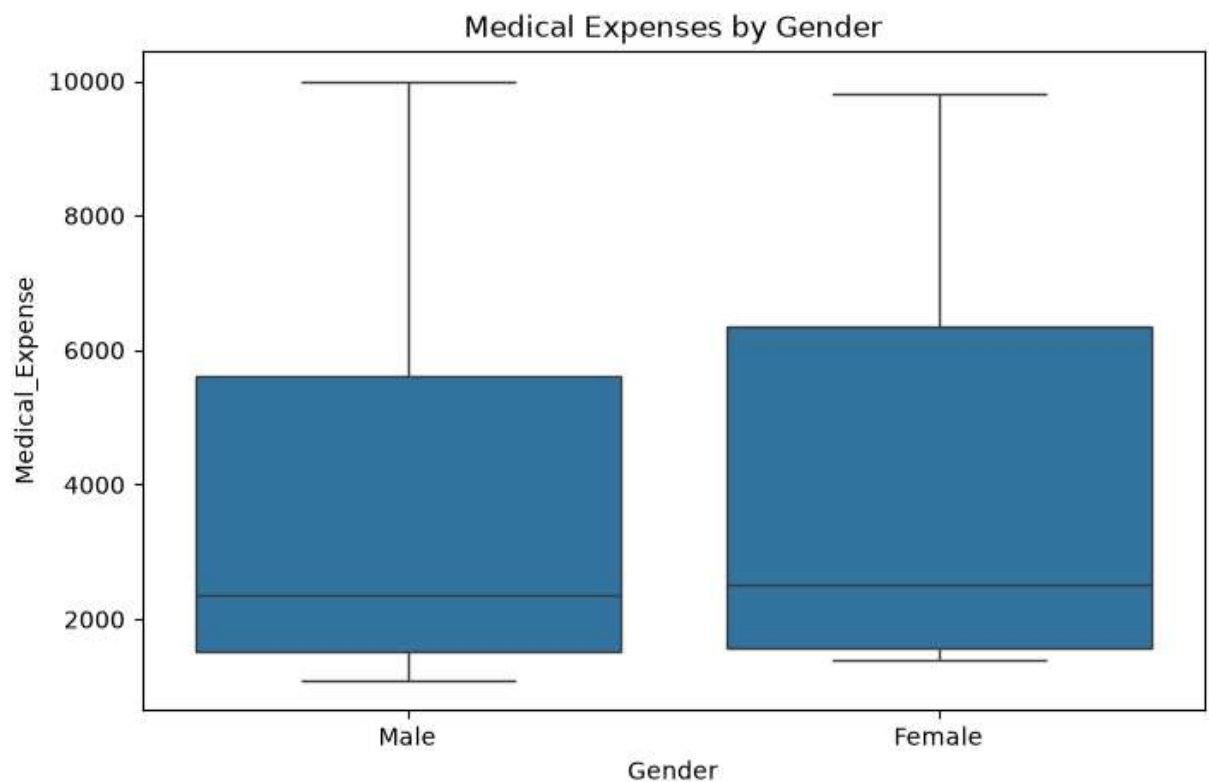
```
In [16]: plt.figure(figsize=(8,5))
sns.boxplot(x="Treatment_Type", y="Medical_Expense", data=df)

plt.title("Medical Expenses by Treatment Type")
plt.show()
```



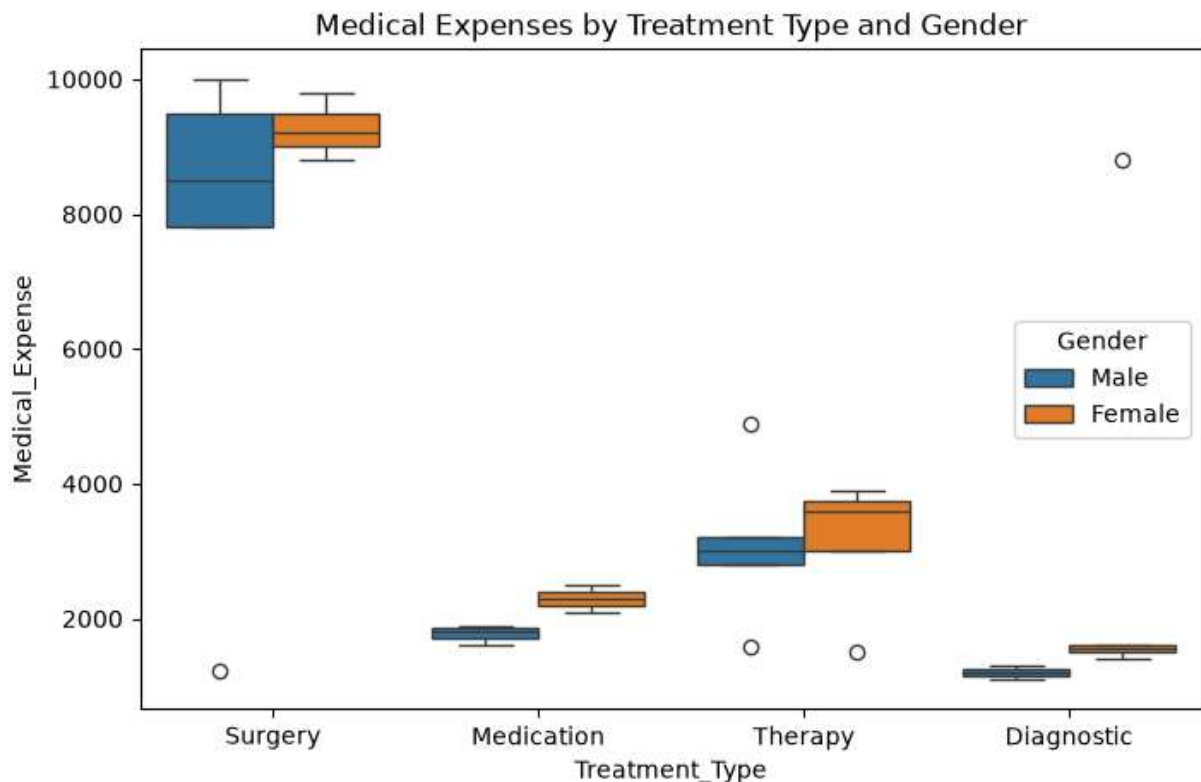
```
In [17]: plt.figure(figsize=(8,5))
sns.boxplot(x="Gender", y="Medical_Expense", data=df)

plt.title("Medical Expenses by Gender")
plt.show()
```



```
In [18]: plt.figure(figsize=(8,5))
sns.boxplot(x="Treatment_Type",
            y="Medical_Expense",
            hue="Gender",
            data=df)

plt.title("Medical Expenses by Treatment Type and Gender")
plt.show()
```



```
In [19]: avg_treatment = df.groupby("Treatment_Type")["Medical_Expense"].mean()

print(avg_treatment)
```

```
Treatment_Type
Diagnostic    2306.500000
Medication    2033.333333
Surgery       8104.375000
Therapy       3119.888889
Name: Medical_Expense, dtype: float64
```

```
In [20]: avg_gender = df.groupby("Gender")["Medical_Expense"].mean()

print(avg_gender)
```

```
Gender
Female    4143.133333
Male      3838.687500
Name: Medical_Expense, dtype: float64
```

```
In [21]: avg_both = df.groupby(["Treatment_Type", "Gender"])["Medical_Expense"].mean()
```

```
print(avg_both)
```

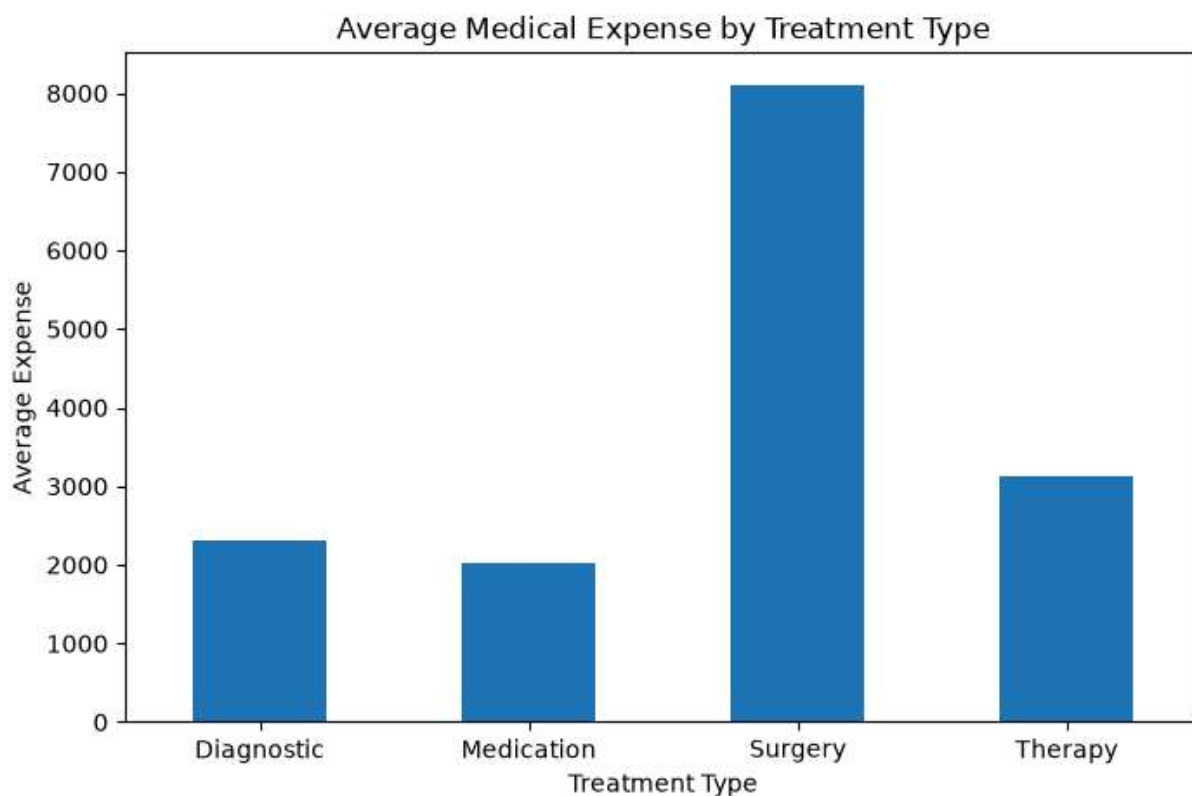
Treatment_Type	Gender	
Diagnostic	Female	2970.400000
	Male	1200.000000
Medication	Female	2300.000000
	Male	1766.666667
Surgery	Female	9266.666667
	Male	7407.000000
Therapy	Female	3148.750000
	Male	3096.800000

Name: Medical_Expense, dtype: float64

```
In [22]: plt.figure(figsize=(8,5))
avg_treatment.plot(kind="bar")

plt.title("Average Medical Expense by Treatment Type")
plt.xlabel("Treatment Type")
plt.ylabel("Average Expense")
plt.xticks(rotation=0)

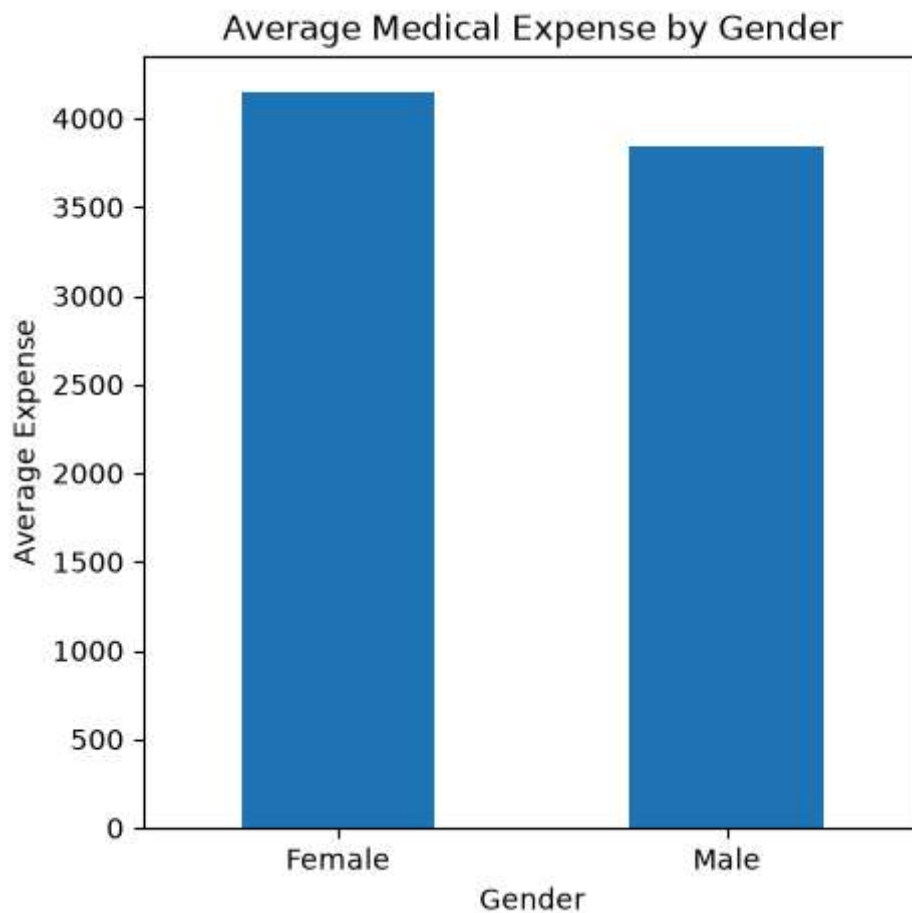
plt.show()
```



```
In [23]: plt.figure(figsize=(5,5))
avg_gender.plot(kind="bar")

plt.title("Average Medical Expense by Gender")
plt.xlabel("Gender")
plt.ylabel("Average Expense")
plt.xticks(rotation=0)
```

```
plt.show()
```



```
In [27]: std_expense = df["Medical_Expense"].std()  
  
print("Standard Deviation =", std_expense)
```

Standard Deviation = 3181.31527306972

```
In [25]: median_expense = df["Medical_Expense"].median()  
  
print("Median Medical Expense =", median_expense)
```

Median Medical Expense = 2500.0

```
In [ ]:
```